

Beyond the Itch: Uncovering Disparities in Atopic Dermatitis Testing Practices

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INTRODUCTION

Atopic dermatitis (AD) is often complicated by allergic contact dermatitis (ACD), for which patch testing is the diagnostic gold standard. However, its use in AD is guided primarily by expert opinion rather than standardized criteria. Variability in clinical practice may delay identification of relevant allergens, hindering treatment optimization. The purpose of this project is to characterize the demographic and clinical profile of the AD population and examine whether disparities exist in access to patch testing.

METHODS

- **Sample & Data Collection:** Single-center retrospective study of 359 de-identified patients with a new diagnosis of AD (June 2024–June 2025) using HERON.
- **Variables analyzed:** Demographics, number of atopic comorbidities, and insurance type.
- **Comparisons:** Demographic and clinical characteristics were compared between patients who received patch testing vs. those who did not.
- **Analysis methods:** Statistical analyses included chi-square tests and t-tests for group comparisons. Logistic regression assessed factors independently associated with testing status.

RESULTS

A total of 359 patients with a new diagnosis of atopic dermatitis were identified between June 2024 and June 2025. Of those, 39 patients received patch testing. Comparisons of demographic characteristics are shown below

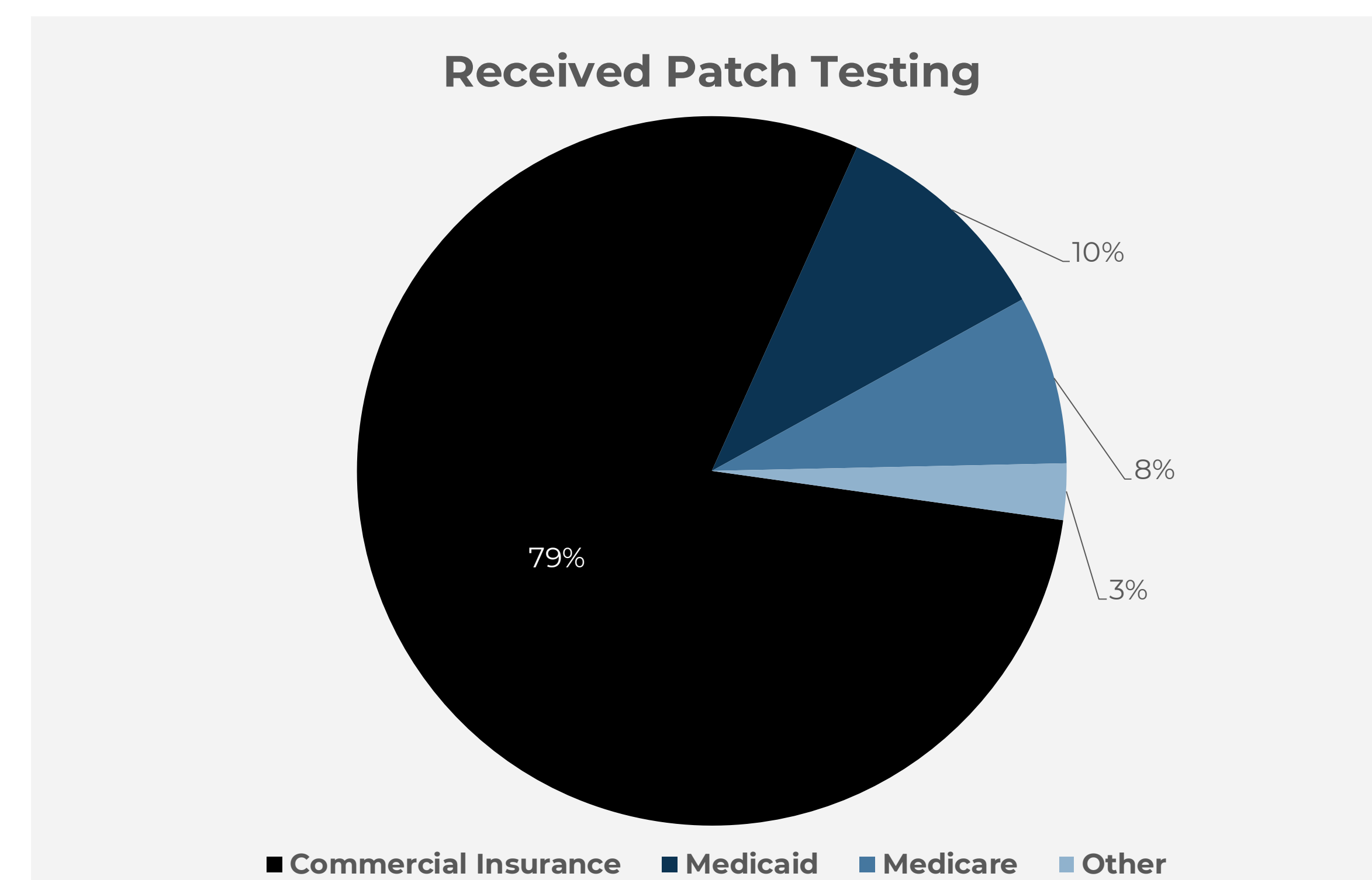


Figure A. Population that received patch testing by insurance type.

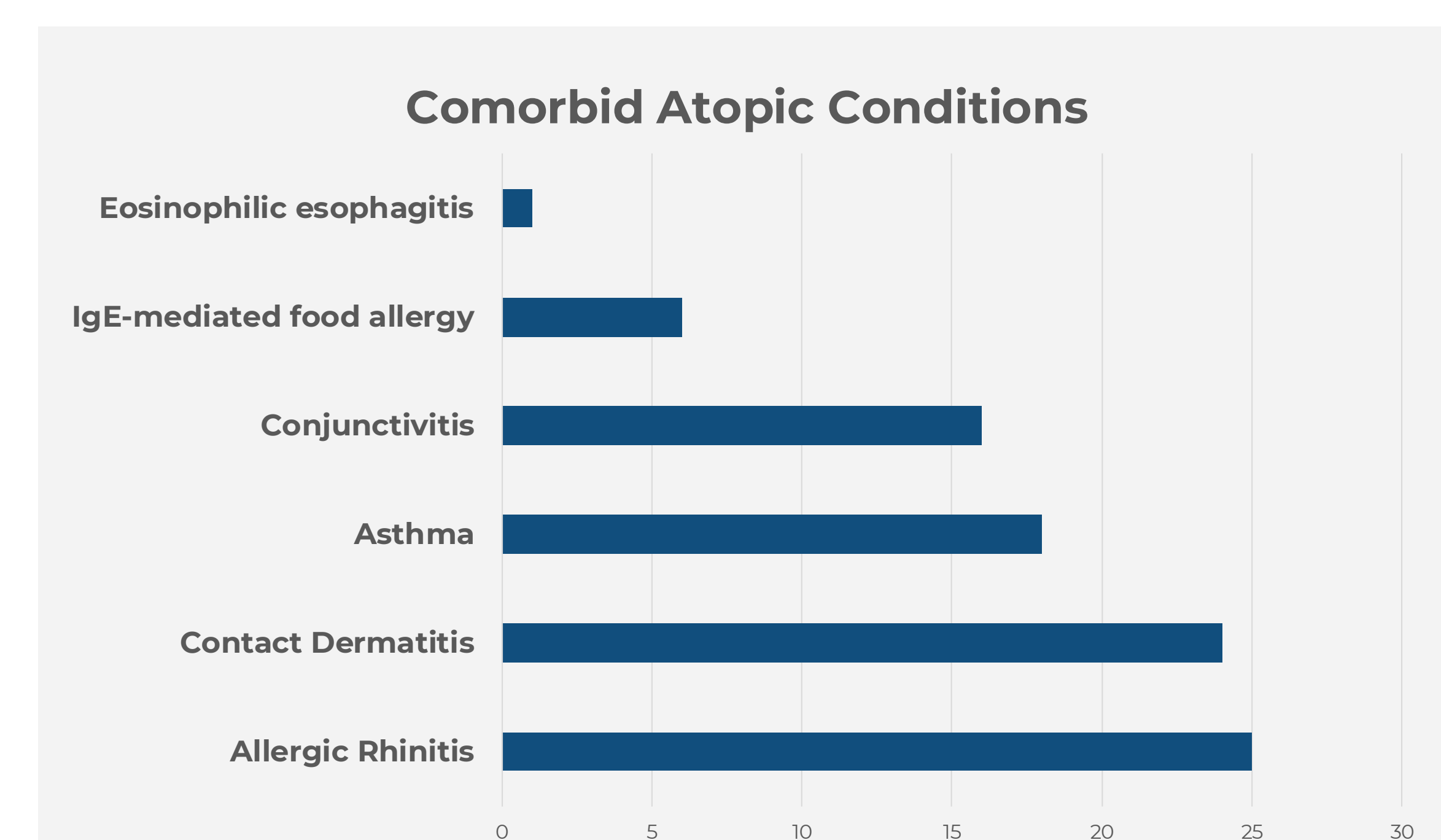


Figure B. Comorbid atopic conditions of those who received patch testing.

Key Predictors of Patch Testing in AD Patients

- **Females** had a 3.4x higher odds of receiving testing ($p < 0.001$)
- Those with **commercial insurance** had a 3.1x higher odds ($p < 0.01$)
- **White patients** were 2x more likely to be tested ($p = 0.041$)
- Mean age = 42 years
+4% higher odds of testing per year of age
- Mean number of **comorbidities** = 2.4
Not independently associated with testing

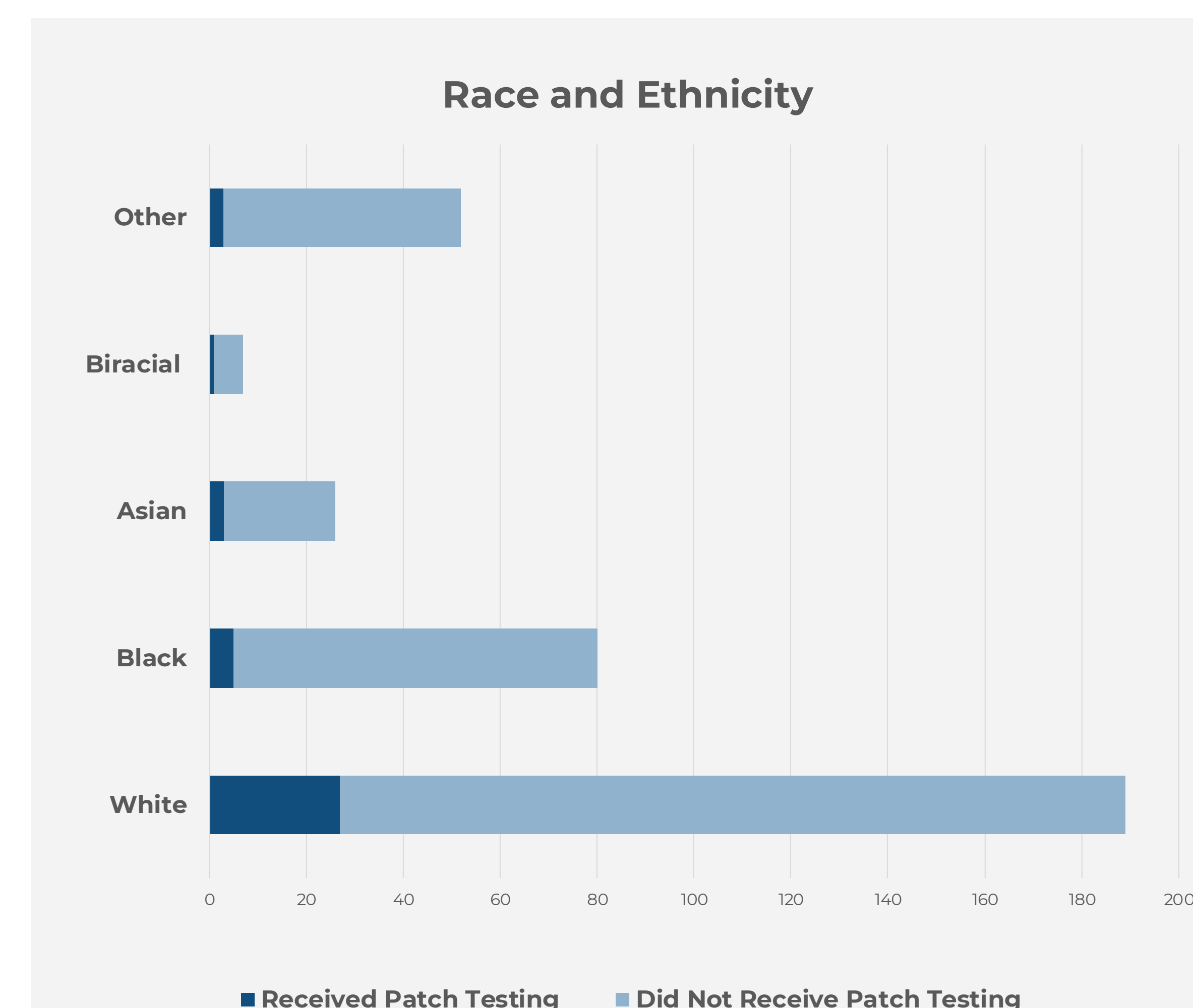


Figure C. Population that received patch testing by race and ethnicity compared to those who did not received patch testing.

DISCUSSION

Patch testing practices for patients with AD remain highly variable, reflecting the lack of standardized guidelines. These inconsistencies may lead to missed evaluations for patients who could benefit from testing and contribute to delays in identifying comorbid allergic contact dermatitis.

Variability in clinician decision-making and patient access may also hinder integration of results into management and create fragmented care. Ongoing expansion of our dataset will help clarify how these patterns influence outcomes and guide more evidence-based, equitable approaches to patch testing in AD.

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